

1. Executive Summary

This comprehensive Hardware Architecture and Implementation Guide provides detailed recommendations for deploying the Djazzy National Security Operations Center (SOC) Platform in a production telecommunications environment. The platform represents a mission-critical infrastructure designed to process over 50 billion security events annually, handle 80 billion Call Detail Records (CDRs) per year, and support monitoring for more than 15 million mobile subscribers across Algeria's largest telecommunications network. The architecture has been engineered specifically for telco-scale operations with stringent requirements for high availability, data integrity, and real-time threat detection capabilities.

The deployment architecture presented in this document addresses three fundamental infrastructure paradigms: on-premises physical server deployment, cloud-based virtualized infrastructure, and hybrid configurations that combine both approaches. Each paradigm offers distinct advantages in terms of capital expenditure versus operational flexibility, data sovereignty compliance, and scalability characteristics. For a national telecommunications operator like Djazzy, where data residency requirements and regulatory compliance are paramount considerations, this guide provides specific recommendations that balance technical performance requirements with operational constraints and budgetary considerations typical of enterprise deployments.

1.1 Platform Resource Requirements Overview

The Djazzy SOC Platform comprises thirty-nine (39) containerized microservices organized across four isolated network segments, each serving specific security functions within the overall operations framework. The aggregate resource requirements for full production deployment have been calculated based on comprehensive analysis of each service's memory footprint, CPU utilization patterns under peak load conditions, and storage I/O characteristics during normal and stress-test scenarios. Understanding these baseline requirements is essential for proper capacity planning and ensures that the deployed infrastructure can maintain service level agreements even during periods of elevated threat activity or network anomalies.

Resource Category	Total Required	Peak Buffer (+20%)	Recommended
RAM Memory	180 GB	216 GB	256 GB (minimum)
vCPU Cores	115 cores	138 cores	160 cores (minimum)
Primary Storage (SSD)	4.5 TB	5.4 TB	8 TB NVMe
Backup Storage (HDD)	15 TB	18 TB	24 TB Enterprise
Network Throughput	100 Gbps	120 Gbps	100 Gbps + 10 Gbps Mgmt

Storage IOPS	150,000+	180,000+	200,000+ NVMe
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Table 1.1: Aggregate Resource Requirements Summary

The resource allocation figures presented above represent conservative estimates derived from extensive load testing and production benchmarking of equivalent security platforms operating at similar scale. The Peak Buffer column incorporates a twenty percent safety margin to accommodate traffic spikes during security incidents, planned maintenance windows with reduced capacity, and organic growth over the initial eighteen-month deployment cycle. Organizations with aggressive subscriber growth projections or plans to expand monitoring scope to additional network segments should consider increasing these baseline figures by an additional fifteen to twenty-five percent to avoid premature capacity exhaustion and costly emergency expansion projects.

2. Hardware Architecture Options

Selecting the appropriate hardware architecture for a national-scale SOC platform requires careful evaluation of multiple factors including total cost of ownership, operational complexity, compliance requirements, scalability trajectory, and organizational expertise in managing different infrastructure paradigms. This section presents three distinct deployment architectures, each optimized for different operational contexts and strategic priorities. The recommendation matrix provided at the end of this section will assist stakeholders in selecting the optimal approach based on their specific constraints and objectives.

2.1 Option A: On-Premises Physical Server Infrastructure

The on-premises deployment model represents the traditional approach to enterprise infrastructure, offering maximum control over hardware resources, network topology, and physical security. For telecommunications operators subject to strict data sovereignty regulations and national security requirements, this approach provides the highest assurance that sensitive security data remains within controlled facilities. The primary trade-off involves higher upfront capital expenditure and longer procurement cycles, though these costs are often offset by lower ongoing operational expenses and complete elimination of recurring cloud subscription fees.

2.1.1 Recommended Server Configuration

For a fully on-premises deployment supporting the complete Djezzy SOC Platform with all thirty-nine services running at production scale, we recommend a distributed cluster architecture consisting of six to eight physical servers arranged in a high-availability configuration with redundant power supplies, multiple network interfaces, and hot-swappable storage components. Each server role has been optimized for the specific workload characteristics of the services it will host, ensuring that compute-intensive operations (such as Elasticsearch indexing and Suricata packet analysis) do not contend with storage-bound workloads (such as PostgreSQL transaction processing and Kafka log segment writes).

Server Role	Quantity	Processor	Memory	Storage	Network
SIEM/Analytics Master Node	2x	Dual Intel Xeon Gold 6348 (28C/56T)	512GB DDR4 3200MHz ECC	2x 960GB NVMe (OS/Logs) + 4x 3.84TB NVMe (RAID 10 Data)	2x 25GbE + 1x 10GbE IPMI
Elasticsearch Data Nodes	3x	Dual Intel Xeon Gold 5318Y (24C/48T)	512GB DDR4 3200MHz ECC	8x 7.68TB NVMe (RAID 10) ~55TB usable	2x 25GbE + 1x 10GbE IPMI

Database Cluster (PostgreSQL/Kafka)	3x	Dual Intel Xeon Silver 4314 (32C/64T)	384GB DDR4 3200MHz ECC	4x 3.84TB NVMe (RAID 10) + 8TB HDD (Archive)	2x 25GbE + 1x 10GbE IPMI
NSM/Package Capture Nodes	2x	Dual Intel Xeon Gold 6248R (20C/40T)	256GB DDR4 3200MHz ECC	2x 960GB NVMe (OS/Cache) + 4x 3.84TB NVMe (PCAP Storage)	2x 100GbE (Capture) + 1x 25GbE (Mgmt)
Application/Orchestration	2x	Single Intel Xeon Gold 6230 (20C/40T)	128GB DDR4 3200MHz ECC	2x 480GB NVMe (RAID 1 OS)	2x 25GbE + 1x 10GbE IPMI

Table 2.1: On-Premises Physical Server Specifications by Role

The server specifications outlined above reflect current-generation Intel Xeon Scalable processors (third generation Ice Lake or newer) which provide excellent performance-per-watt ratios and robust virtualization support through Intel VT-x and VT-d technologies. AMD EPYC processors (Milan or Genoa generations) represent viable alternatives that may offer better core density and memory channel bandwidth for certain workload profiles, particularly the Elasticsearch data nodes which benefit from increased memory throughput. All servers should be specified with redundant hot-swap power supplies (N+1 configuration), hot-plug cooling fans, and at least two independent power feeds connected to separate uninterruptible power supply (UPS) units with automatic transfer switch capability.

2.1.2 Storage Architecture Details

Storage subsystem design represents one of the most critical aspects of SOC platform performance, as security event processing generates extremely write-intensive workloads with unpredictable access patterns. The recommended configuration utilizes NVMe solid-state drives exclusively for all active data volumes, with enterprise SATA HDDs reserved only for long-term archival storage with relaxed performance requirements. For Elasticsearch clusters, we recommend dedicated NVMe drives configured in RAID 10 arrays to provide both redundancy and improved sequential write performance, which directly impacts index refresh latency and query response times during incident investigations.

Service Component	Storage Type	Capacity	IOPS Requirement	Latency Target
Elasticsearch Indices	NVMe RAID 10	~50 TB usable	100,000+ random write	<2ms p99 write
PostgreSQL Database	NVMe RAID 10	~4 TB usable	25,000+ random R/W	<1ms p99 read
Kafka Event Log	NVMe RAID 10	~8 TB usable	75,000+ sequential write	<5ms p99 append
PCAP Capture (Suricata/Zenith)	NVMe RAID 10	~12 TB usable	50,000+ sequential write	<10ms capture
Redis Cache Layer	NVMe RAID 1	~500 GB usable	200,000+ random R/W	<0.5ms p99 read

Backup Archive	Enterprise HDD	~20 TB usable	2,000+ sequential	N/A (batch)
OS/System Volumes	NVMe RAID 1	~2 TB total	Standard	<1ms read

Table 2.2: Detailed Storage Requirements by Service Component

2.2 Option B: Cloud Infrastructure (Private Cloud / VM-Based)

Cloud-based deployment offers significant advantages in terms of deployment speed, operational flexibility, and ability to scale resources dynamically in response to changing demand patterns. For organizations with existing private cloud infrastructure or established relationships with Infrastructure-as-a-Service providers, this option can substantially reduce time-to-production while maintaining acceptable performance characteristics for most SOC workloads. However, cloud deployment introduces dependencies on external network connectivity and requires careful attention to data classification policies and regulatory compliance frameworks governing security data residence.

2.2.1 Virtual Machine Specifications

VM Role	vCPUs	Memory	Storage	Instance Type Equivalent	Quantity
SIEM Master (Wazuh/Elasticsearch Master)	32 vCPU	128 GB RAM	1 TB GP3/IO1	r5.4xlarge / Standard_E32s_v3	2
Elasticsearch Data Node	32 vCPU	256 GB RAM	4 TB IO1 Optimized	r5.8xlarge / Standard_E64s_v3	4
PostgreSQL Primary	16 vCPU	64 GB RAM	1 TB Provisioned IOPS	r5.4xlarge / Standard_E32s_v3	1
PostgreSQL Replica	16 vCPU	64 GB RAM	1 TB Provisioned IOPS	r5.4xlarge / Standard_E32s_v3	3
Kafka Broker	16 vCPU	64 GB RAM	2 TB Throughput Optimized	i3.4xlarge / Standard_L32s_v3	3
Kafka ZooKeeper	4 vCPU	16 GB RAM	100 GB SSD	t3.xlarge / Standard_F4s_v2	3
NSM Engine (Suricata/Zeek)	32 vCPU	128 GB RAM	2 TB Enhanced Network	r5.4xlarge / Standard_F32s_v2	2
PCAP Analysis (Arkime)	16 vCPU	64 GB RAM	3 TB Storage Optimized	i3.4xlarge / Standard_L16s_v3	2
SOAR Platform (TheHive/Cortex)	8 vCPU	32 GB RAM	500 GB GP3	r5.2xlarge / Standard_E8s_v3	2
Threat Intel (MISP/OpenCTI)	8 vCPU	32 GB RAM	500 GB GP3	r5.2xlarge / Standard_E8s_v3	2
Vulnerability (OpenVAS/DefectDojo)	8 vCPU	32 GB RAM	500 GB GP3	r5.2xlarge / Standard_E8s_v3	2
Monitoring (Prometheus/Grafana)	8 vCPU	32 GB RAM	500 GB GP3	r5.2xlarge / Standard_E8s_v3	2
API Gateway (Kong)	4 vCPU	16 GB RAM	100 GP3	t3.xlarge / Standard_F4s_v2	2
Frontend Application	8 vCPU	32 GB RAM	100 GB GP3	t3.2xlarge / Standard_F8s_v2	2

Table 2.3: Virtual Machine Specifications for Cloud Deployment

The virtual machine specifications above map to commonly available instance types from major cloud providers including Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). Instance types have been selected to optimize the balance between compute capacity, memory bandwidth, and storage I/O performance required by each workload category. For production deployments, we strongly recommend utilizing dedicated host instances or dedicated tenancy options to ensure resource isolation and prevent noisy neighbor effects from impacting SOC platform performance during critical security operations. Reserved instance commitments can reduce costs by forty to sixty percent compared to on-demand pricing for steady-state workloads with predictable resource consumption patterns.

2.3 Option C: Hybrid Architecture (Recommended)

The hybrid architecture combines the strengths of both on-premises and cloud deployment models, positioning latency-sensitive and data-intensive workloads on local infrastructure while leveraging cloud resources for elastic scaling, disaster recovery, and non-critical processing tasks. This approach is particularly well-suited for telecommunications operators who must maintain continuous operation capability even during wide-area network outages or cloud provider incidents, while still benefiting from cloud economics for development, testing, and burst capacity scenarios.

In the recommended hybrid configuration, the core SIEM pipeline (Wazuh managers, Elasticsearch hot nodes, Kafka brokers, and PostgreSQL primary database) resides on-premises to ensure zero-latency event ingestion and immediate availability for real-time alerting. Elasticsearch warm and cold tiers, backup archives, development/testing environments, and analytical workloads that tolerate higher latency operate in the cloud region, taking advantage of virtually unlimited storage scalability and pay-per-use pricing models. This tiered approach optimizes both performance characteristics and total cost of ownership while maintaining operational resilience against various failure modes.

Component Category	On-Premises	Cloud	Rationale
Real-time Event Ingestion	100%	0%	Latency sensitivity, network independence
Elasticsearch Hot Tier (7 days)	100%	0%	Query performance, investigation SLAs
Elasticsearch Warm Tier (30 days)	0%	100%	Cost optimization, infrequent access
Elasticsearch Cold Tier (>30 days)	0%	100%	Archive storage economics
Database Primary (PostgreSQL)	100%	0%	Transaction consistency, low latency
Database Replicas	50%	50%	Read scaling, geographic distribution
Kafka Brokers (Hot Topics)	100%	0%	Real-time stream processing

Kafka Brokers (Archive Topics)	0%	100%	Long-term retention, replay
Development/Test Environments	0%	100%	Cost efficiency, isolation
Disaster Recovery Site	0%	100%	Geographic separation, rapid provisioning
Backup Storage	0%	100%	Durability, immutability, cost

Table 2.4: Hybrid Architecture Workload Distribution Strategy

3. Network Infrastructure Requirements

Network infrastructure design for a national-scale SOC platform requires careful attention to throughput capacity, latency characteristics, segmentation boundaries, and monitoring visibility. The Djezzy SOC Platform processes network traffic at rates up to 100 gigabits per second during peak periods, necessitating purpose-built network fabrics capable of sustaining line-rate packet capture without loss while simultaneously supporting east-west traffic between containerized services and north-south traffic to analyst workstations and external integration points.

3.1 Network Segmentation Architecture

The platform implements four distinct network isolation zones, each serving specific functional purposes with controlled traffic flows between zones via firewall rules and API gateway policies. This segmentation strategy limits blast radius of potential compromises, enables granular access control policies, and simplifies compliance auditing by providing clear boundary definitions for different data classifications and trust levels. Each zone operates on dedicated VLANs or overlay networks with IP address ranges allocated from private RFC1918 address space according to the following scheme defined in the production Docker Compose orchestration layer.

Network Zone	CIDR Range	Purpose	Connected Services	Bandwidth
soc-frontend	172.28.0.0/16	User-facing applications, API Gateway, Web UI	Next.js App, Kong API GW, Grafana Dashboards	10 Gbps
soc-backend	172.29.0.0/16	Core services, SOAR, Threat Intel	TheHive, Cortex, MISP, OpenCTI, DefectDojo	25 Gbps
soc-events	172.30.0.0/16	Event processing, SIEM, Analytics	Wazuh, Elasticsearch, Kafka, PostgreSQL	100 Gbps
soc-monitoring	172.31.0.0/16	Observability, Metrics, Logging	Prometheus, AlertManager, Log Aggregation	10 Gbps

Table 3.1: Network Zone Configuration and Service Assignment

3.2 Physical Network Interface Requirements

Each physical server in the deployment requires multiple network interfaces to support the segmented architecture while maintaining management access and out-of-band administration capabilities. The following interface layout assumes industry-standard 1U and 2U rack-mountable servers with available PCIe slots for additional network interface cards where onboard ports prove insufficient. Network interface controllers should support Single Root I/O Virtualization (SR-IOV) to enable direct VM-to-NIC passthrough for packet capture workloads, and Data Plane Development Kit (DPDK) compatibility for

userspace packet processing in Suricata and Zeek engines.

Server Role	Data Interfaces	Management	IPMI/iDRAC	Total Ports	Recommended NIC
SIEM/Analytics Master Node	2x 25GbE SFP28	1x 10GbE BASE-T	1x 1GbE dedicated	4	Intel E810-CAM2 or Mellanox CX5
Elasticsearch Data Node	2x 25GbE SFP28	1x 10GbE BASE-T	1x 1GbE dedicated	4	Intel E810-CAM2 or Mellanox CX5
Database Cluster Node	2x 25GbE SFP28	1x 10GbE BASE-T	1x 1GbE dedicated	4	Intel E810-XXV or Broadcom P7
NSM/Packet Capture Node	2x 100GbE QSFP28	1x 25GbE SFP28	1x 1GbE dedicated	4	Mellanox ConnectX-6 or Napatech SmartNIC
Application/Orchestration Node	2x 25GbE SFP28	1x 10GbE BASE-T	1x 1GbE dedicated	4	Intel X710-DA2 (onboard sufficient)

Table 3.2: Network Interface Card Requirements by Server Role

3.3 Switch Fabric and Topology

The network switching fabric forms the backbone of SOC platform connectivity and must be designed with redundancy, low latency, and adequate buffer capacity for bursty security event traffic. We recommend a spine-leaf architecture using modern data center switches with support for lossless Ethernet (via Priority Flow Control and Data Center Bridging protocols) to prevent packet loss during congestion events that could otherwise result in missed security detections or incomplete forensic captures. All inter-switch links should utilize 100 Gigabit Ethernet to provide ample headroom for east-west traffic growth as the platform scales to accommodate additional data sources and analytical workloads.

For the recommended twelve-server deployment (including application nodes), a minimum of two spine switches and four leaf switches provides full bisectional bandwidth with N+1 redundancy at every layer. Spine switches should be chassis-based modular platforms with line cards providing at least forty-eight 100GbE ports and support for VXLAN and EVPN overlay protocols if multi-tenancy or stretched VLAN requirements emerge in future phases. Leaf switches may be fixed-configuration 1U or 2U platforms with twenty-four to forty-eight 25GbE server-facing ports and four to eight 100GbE uplink ports connecting to the spine layer. All switches should support IEEE 1588 Precision Time Protocol (PTP) for accurate timestamp correlation across distributed capture nodes.

Switch Role	Model Example	Port Configuration	Throughput	Quantity
Spine Switch	Cisco Nexus 9504 Arista 7800R3 Juniper QFX10002	48x 100GbE + 6x 400GbE	57.6 Tbps	2 (redundant)

Leaf Switch (Computer)	Cisco Nexus 9364C Arista 7050X3 Juniper QFX5110	48x 25GbE + 8x 100GbE	6.4 Tbps	2-3
Leaf Switch (Storage)	Cisco Nexus 9364C Arista 7050X3 Juniper QFX5110	48x 25GbE + 8x 100GbE	6.4 Tbps	1-2
Management Switch	Cisco Catalyst 9300 Arista 7050X Juniper EX4300	48x 1GbE/10GbE + 4x 25GbE uplink	480 Gbps	2 (redundant)

Table 3.3: Recommended Switch Infrastructure Components

4. High Availability and Redundancy Design

Mission-critical security operations require infrastructure designs that maintain service continuity through component failures, maintenance windows, and unexpected outage scenarios. The Djezzy SOC Platform achieves high availability through redundant deployment of all critical services, automated failover mechanisms, and geographically distributed disaster recovery capabilities. This section details the availability targets, redundancy strategies, and operational procedures necessary to meet the ninety-nine point nine five percent (99.95%) uptime commitment required for national telecommunications security operations.

4.1 Availability Targets by Service Tier

Service Tier	Availability Target	Max Downtime/Year	Recovery Objective	Redundancy Mode
Event Ingestion Pipeline	99.99% (~52 min)	52.56 minutes	RTO < 5 min RPO < 1 min	Active-Active
Real-time Alerting	99.99% (~52 min)	52.56 minutes	RTO < 2 min RPO < 30 sec	Active-Active
SIEM Search/Investigation	99.95% (~4.4 hr)	4.38 hours	RTO < 15 min RPO < 5 min	Active-Passive
Case Management (SOAR)	99.9% (~8.8 hr)	8.76 hours	RTO < 30 min RPO < 15 min	Active-Passive
Threat Intelligence Feeds	99.9% (~8.8 hr)	8.76 hours	RTO < 1 hour RPO < 1 hour	Active-Passive
Vulnerability Scanning	99.5% (~44 hr)	43.8 hours	RTO < 4 hours RPO < 24 hours	N+1 Capacity
Reporting/Analytics	99.5% (~44 hr)	43.8 hours	RTO < 4 hours RPO < 24 hours	N+1 Capacity

Table 4.1: Service-Level Availability Targets and Recovery Objectives

4.2 Cluster Topology and Failover Mechanisms

Each critical service in the Djezzy SOC Platform deploys in clustered configurations with automatic failover capabilities. Elasticsearch operates as a dedicated cluster with a minimum of three master-eligible nodes to prevent split-brain scenarios, with data distributed across replicas using awareness attributes that ensure replica placement on different fault domains (racks, power zones, or availability zones). PostgreSQL uses streaming replication with synchronous commit mode for the primary-replica relationship, ensuring zero data loss upon failover at the cost of slightly increased write latency. Kafka

maintains an in-sync replica set (ISR) of at least two brokers per partition, with leader election coordinated through ZooKeeper or KRaft consensus depending on the deployed version.

4.3 Disaster Recovery Strategy

Beyond local high availability mechanisms, the SOC platform requires a geographically separated disaster recovery site capable of assuming full production operations within the defined recovery time objective. The recommended approach employs asynchronous replication for most data stores, accepting a recovery point objective of up to fifteen minutes for non-real-time services while maintaining synchronous or near-synchronous replication for the event ingestion pipeline where data loss is unacceptable. The DR site should be located at least one hundred kilometers from the primary data center to provide protection against regional disasters such as earthquakes, floods, or wide-area power grid failures, while remaining within acceptable network latency bounds for database replication traffic.

Component	Replication Method	RPO Target	Failover Mechanism	Validation Frequency
Elasticsearch Indices	Cross-Cluster Replication (CCR)	< 15 minutes	Promote remote cluster	Monthly DR drill
PostgreSQL Database	Streaming Replication (async)	< 5 minutes	pg_ctl promote trigger	Quarterly failover test
Kafka Topics	MirrorMaker 2 / Replicator	< 10 minutes	Consumer group offset sync	Monthly replication lag check
Configuration (GitOps)	Git repository mirror	< 1 minute	Automatic Git pull	Continuous (CI/CD)
Container Images	Registry mirroring	< 30 minutes	Local registry failover	Weekly image sync validation
Secrets/Credentials	Vault replication / HSM	< 5 minutes	Standby Vault activation	Quarterly secrets rotation

Table 4.2: Disaster Recovery Replication Strategy by Component

5. Implementation Phases and Timeline

Successful deployment of a platform with the complexity and scale of the Djezzy SOC System requires a structured implementation approach that manages risk through incremental validation, builds operational expertise progressively, and delivers measurable value at each milestone. This section outlines a six-phase implementation roadmap spanning approximately nine months from project kickoff to full production operation, with defined deliverables, success criteria, and go/no-go decision points at each phase boundary. Organizations with existing infrastructure or prior experience with similar platforms may compress certain phases, but the logical sequencing of activities should remain intact to prevent rework and ensure stable foundation layers before building dependent capabilities.

5.1 Phase 1: Infrastructure Preparation (Weeks 1-4)

The initial phase focuses on establishing the physical and virtual foundation upon which all subsequent platform components depend. Activities include procurement and rack mounting of server hardware, installation and cabling of network switching fabric, power and cooling verification, and base operating system deployment across all nodes. This phase also establishes the container runtime environment (Docker Enterprise or Podman with compatible orchestrator), configures network bridges and VLAN trunks according to the segmentation design, and validates basic inter-node connectivity at expected throughput levels. By phase end, the infrastructure team should demonstrate successful ping connectivity between all nodes on all VLANs, validated IPMI/iDRAC access for remote management, and confirmed power redundancy through simulated feed failure tests.

- Complete hardware procurement and receive all server, storage, and networking equipment
- Install equipment in designated racks with proper cable management and labeling
- Configure UPS systems and validate automatic transfer switch operation
- Deploy base operating system (Rocky Linux 9 / Ubuntu 22.04 LTS) on all nodes
- Install container runtime (Docker CE 27.x / Podman 5.x) and configure daemon.json
- Establish network bridges for soc-frontend, soc-backend, soc-events, soc-monitoring VLANs
- Validate network throughput meets specifications (iperf3 testing at line rate)
- Configure centralized logging (rsyslog/journald forwarding) and NTP synchronization
- Document all IP addresses, MAC addresses, serial numbers, and asset tags in CMDB
- Conduct physical security audit and verify environmental monitoring sensors

5.2 Phase 2: Core Platform Deployment (Weeks 5-8)

Phase two introduces the foundational platform services that enable all subsequent security tooling. This includes deploying the PostgreSQL database cluster with streaming

replication, establishing the Kafka message broker cluster for event streaming, initializing the Elasticsearch cluster with appropriate index templates and lifecycle policies, and configuring Redis cache instances for session management and temporary data stores. The Docker Compose orchestration layer is deployed and validated, demonstrating successful container scheduling across the infrastructure with proper network attachment and volume mount behavior. This phase concludes with smoke tests confirming database connectivity, message broker publish-subscribe functionality, and basic Elasticsearch index and search operations.

- Deploy PostgreSQL 16 cluster (1 primary, 2 replicas) with PgBouncer connection pooling
- Initialize Prisma ORM schema migrations and validate all 15 enterprise data models
- Deploy Apache Kafka 3.6 cluster (3 brokers) with ZooKeeper or KRaft controller
- Create Kafka topics with appropriate partition counts and replication factors
- Deploy Elasticsearch 8.12 cluster (3 master + 3 data nodes minimum)
- Configure Elasticsearch index templates, ILM policies, and snapshot repositories
- Deploy Redis 7 cluster (sentinel mode) for caching and session state
- Deploy Kong API Gateway with SSL termination and rate limiting policies
- Validate docker-compose.prod.yml successfully starts all infrastructure services
- Execute load tests to confirm baseline performance meets specifications

5.3 Phase 3: Security Tool Integration (Weeks 9-14)

Phase three encompasses the deployment and integration of the fifteen open-source security tools that comprise the Djezzy SOC analytical capabilities. Tools are deployed in dependency order, beginning with the SIEM layer (Wazuh, Elasticsearch, Kibana) that receives and indexes all security events, followed by the network security monitoring suite (Suricata, Zeek, Arkime) that captures and analyzes network traffic, then endpoint detection and response components (GRR, Osquery Fleet), threat intelligence platforms (MISP, OpenCTI), security orchestration and automation (TheHive, Cortex), vulnerability management (OpenVAS/GVM, DefectDojo), and finally the observability stack (Prometheus, Grafana). Each tool deployment includes configuration tuning for telco-scale data volumes, integration testing with upstream and downstream components, and documentation of operational procedures for common administrative tasks.

Week	Tools Deployed	Integration Points	Validation Criteria
9-10	Wazuh SIEM, Elasticsearch, Kibana	Agent registration, index creation	Events indexed, dashboards render
10-11	Suricata IDS/IPS, Zeek NSM	EVE JSON to Kafka, logs to ES	Alerts generated, PCAP captured
11-12	Arkime PCAP analysis, GRR EDR	Kafka consumer, API integration	Full packet search, remote forensics
12-13	Osquery Fleet, TheHive SOAR, Cortex	Flex server, case APIs, analytics	Endpoint queries, case workflows

13-14	MISP, OpenCTI, OpenVAS, DefectDojo	Threat feeds, scan results, findings	IOC enrichment, vuln reports
14	Prometheus, Grafana, AlertManager	Service discovery, dashboarding	Malware collected, alerts fired

Table 5.1: Security Tool Deployment Sequence and Validation Milestones

5.4 Phase 4: Application Layer and Frontend (Weeks 15-17)

With foundational services and security tools operational, phase four deploys the custom Next.js 16 frontend application and backend API services that provide the unified operator experience. This includes the React 19 user interface with shadcn/ui components, the TypeScript backend services implementing telecom-specific business logic (fraud detection algorithms, CDR correlation engine, subscriber profiling), and the integration coordinator that orchestrates cross-tool workflows. This phase also implements the authentication and authorization framework (LDAP integration, role-based access control, audit logging) that was intentionally deferred from earlier phases to allow unrestricted access during integration troubleshooting. User acceptance testing begins during this phase with pilot user groups from the security operations team providing feedback on workflow usability and feature completeness.

5.5 Phase 5: Testing and Hardening (Weeks 18-22)

Phase five focuses on comprehensive testing, performance optimization, and security hardening of the integrated platform. Activities include load testing at projected production volumes (validating 50K EPS sustained ingestion, concurrent user loads, query response times under index pressure), penetration testing of the application layer and API endpoints, failure mode testing (simulated node failures, network partitions, storage degradation), and security configuration review (TLS certificate validation, cipher suite hardening, header security). Runbook development proceeds in parallel, documenting standard operating procedures for incident response, escalation procedures, and disaster recovery execution. This phase culminates in a seven-day production simulation exercise that exercises all operational procedures under realistic conditions without affecting live network monitoring.

5.6 Phase 6: Production Cutover and Hypercare (Weeks 23-26)

The final phase transitions the platform from pre-production to live operation, beginning with data migration from legacy systems (if applicable), gradual traffic shifting to route live security events through the new platform, and formal handoff from project team to operations team. The hypercare period (weeks 24-26) maintains augmented staffing with developers and engineers on standby to rapidly address any issues discovered under genuine production load. Daily standups assess system health, address operator feedback,

and prioritize any bug fixes or usability improvements identified during initial production operation. Upon successful completion of hypercare, the project transitions to steady-state operations with standard change management processes and quarterly business reviews assessing platform effectiveness against defined security metrics and key performance indicators.

Phase	Duration	Key Deliverable	Go/No-Go Criteria
1. Infrastructure Prep	Weeks 1-4	Operational bare-metal cluster	All nodes reachable, network validated
2. Core Platform	Weeks 5-8	DB, messaging, search online	Services healthy, smoke tests pass
3. Security Tools	Weeks 9-14	15 tools integrated	All tools receiving/processing data
4. Application Layer	Weeks 15-17	Frontend, API, auth ready	UAT sign-off from pilot users
5. Testing/Hardening	Weeks 18-22	Validated, hardened platform	Load tests pass, pen test clean
6. Cutover/Hypercare	Weeks 23-26	Production operation	7-day stability, ops team trained

Table 5.2: Implementation Phase Summary and Success Gates

6. Cost Estimation and Budget Planning

Accurate cost estimation for SOC platform deployment requires consideration of both capital expenditures (hardware procurement, software licensing, professional services) and operational expenditures (power, cooling, facility costs, ongoing maintenance, personnel). This section provides order-of-magnitude cost ranges for each deployment option discussed earlier, recognizing that actual costs will vary significantly based on geographic location, vendor relationships, existing infrastructure that can be leveraged, and organizational-specific requirements that may increase or decrease scope relative to the baseline platform specification.

6.1 On-Premises Deployment Cost Breakdown

Category	Item Description	Specification	Unit Cost (USD)	Qty	Total (USD)
Compute Servers	SIEM/Analytics Master x2	Dual Xeon 6348, 512GB, 100TB NVMe 2	\$45,000	2	\$90,000
	Elasticsearch Data x3	Dual Xeon 5318Y, 512GB, 100TB NVMe 3	\$55,000	3	\$165,000
	Database Cluster x3	Dual Xeon 4314, 384GB, 140TB NVMe 3	\$42,000	3	\$126,000
	NSM Packet Capture x2	Dual Xeon 6248R, 256GB, 100TB NVMe 2	\$48,000	2	\$96,000
	Application Servers x2	Single Xeon 6230, 128GB, 100TB NVMe 2	\$18,000	2	\$36,000
Network Infrastructure	Spine Switches x2	100GbE, 48-port modular	\$85,000	2	\$170,000
	Leaf Switches x4	25GbE/100GbE, 48-port	\$35,000	4	\$140,000
	Management Switches x2	10GbE, 48-port L3	\$12,000	2	\$24,000
	Network Cabling & Optics	DAC, AOC, fiber, patch-panels		1	\$25,000
Storage Expansion	NVMe Drives (spare)	7.68TB enterprise U.3	\$1,800	6	\$10,800
	HDD Archive Storage	16TB enterprise SAS	\$450	12	\$5,400
Infrastructure	UPS Systems	20kVA double-conversion	\$15,000	2	\$30,000
	PDU & Power Distribution	Switched, monitored PDU	\$3,500	6	\$21,000
	Rack Enclosures	42U closed, perforated	\$4,500	3	\$13,500
Software & Licensing	OS Licenses (RHEL/Rocky)	Annual subscription	\$2,500	12	\$30,000
	Container Runtime	Docker EE / Podman support	Annual	-	\$25,000
Professional Services	Deployment & Integration	Implementation support		1	\$120,000
	Training & Knowledge Transfer	Operator certification program		1	\$35,000

CONTINGENCY (15% risk buffer for overruns)					\$261,315
GRAND TOTAL					\$2,008,015

Table 6.1: On-Premises Deployment Capital Expenditure Estimate

The cost estimate presented above reflects list pricing for enterprise-grade hardware from major vendors (Cisco, Dell EMC, HPE, Supermicro) and assumes single-site deployment without geographic redundancy. Organizations with established vendor agreements or willingness to consider original design manufacturer (ODM) servers (such as Quanta, Wiyynn, or Inspur) can achieve twenty to forty percent reduction in compute server costs while maintaining comparable performance characteristics. Software costs assume commercial support subscriptions for operating systems and container runtimes; organizations comfortable with community-supported alternatives (Rocky Linux, upstream Docker CE) can eliminate these line items entirely, though this shifts cost burden to internal expertise requirements.

6.2 Annual Operational Expenditure

Category	Item	Annual Cost (USD)	Notes
Facility Costs	Power consumption (12 servers @ avg \$800/W)	\$9,600,000	24/7 operation, \$0.10/kWh
	Cooling (additional HVAC load ~100kW)	\$12,000,000	Estimated 1.5x PUE factor
	Colocation/rack space (if not owned DC)	\$48,000	3 full racks @ \$1,500/rack/mo
Maintenance	Hardware maintenance contracts (12% of HW)	\$1,100,000	4-hour response, parts included
	Software support renewals	\$55,000	OS, container runtime, optional tools
Personnel	SOC Platform Engineers (3 FTE)	\$210,000	\$70k average fully-loaded
	Security Analysts (operations, not platform)	\$0	Separate budget line
Consumables	Storage media replacement (annualized)	\$15,000	Drive failures, capacity growth
	Network equipment sparing	\$8,000	Transceivers, cables, spare parts
Training & Certification	Annual training budget	\$25,000	Conferences, certifications, courses
ANNUAL OPEX TOTAL		\$661,000	Excludes analyst personnel

Table 6.2: Estimated Annual Operational Expenditure

6.3 Cloud Deployment Cost Comparison

Cloud deployment eliminates upfront capital expenditure in favor of ongoing operational costs that scale with actual resource consumption. Based on the virtual machine

specifications outlined in Section 2.2 and current public cloud pricing (us-east-1 / West Europe regions), estimated monthly costs for equivalent cloud deployment range from eighty thousand to one hundred twenty thousand USD monthly depending on instance types selected, committed use discount levels, and actual storage consumption patterns. Over a five-year total cost of ownership horizon, cloud deployment typically costs thirty to fifty percent more than on-premises deployment for steady-state workloads like SOC platforms, though it offers superior flexibility for organizations with variable demand patterns or limited access to data center facilities.

Cost Category	On-Premises (5yr TCO)	Cloud (5yr TCO)	Hybrid (5yr TCO)
Initial CapEx	\$2,008,015	\$0 (migration only)	\$1,200,000
Annual OpEx (avg)	\$661,000/year	\$1,080,000/year	\$780,000/year
5-Year Total	\$5,313,015	\$5,400,000	\$5,100,000
Monthly Equivalent	\$88,550	\$90,000	\$85,000
Advantages	Lowest long-term cost, data sovereignty	Fastest deployment, elastic scaling	Balanced cost/flexibility, DR built-in
Best For	Mature orgs with DC access, steady workloads	Rapid deployment needs, variable demand	Most enterprises, risk-balanced approach

Table 6.3: Five-Year Total Cost of Ownership Comparison by Deployment Model

7. Recommendations and Next Steps

7.1 Recommended Deployment Approach

Based on comprehensive analysis of technical requirements, operational constraints, cost factors, and risk considerations specific to national telecommunications security operations, we recommend the ****Hybrid Architecture (Option C)**** as the optimal deployment approach for the Djazzy SOC Platform. This recommendation balances the need for low-latency, high-throughput event processing with data sovereignty requirements inherent to Algerian telecommunications operations, while preserving flexibility to leverage cloud economics for non-critical workloads and disaster recovery capabilities. The hybrid approach also provides a natural migration path for organizations currently operating legacy on-premises security tools who wish to modernize incrementally rather than executing a high-risk big-bang cutover.

Specifically, we recommend deploying the real-time event ingestion pipeline (packet capture, Kafka hot topics, Elasticsearch hot tier), primary database infrastructure, and core SIEM processing on physical servers located in Djazzy's primary data center facilities. This ensures that even complete loss of external connectivity does not interrupt security monitoring capabilities, a critical requirement for national telecommunications infrastructure protection. Simultaneously, we recommend establishing cloud presence (preferably in a region with strong connectivity to Algeria such as EU-West or Middle East regions) for Elasticsearch warm/cold tiers, development and testing environments, backup storage, and disaster recovery standby infrastructure.

7.2 Immediate Action Items

Week 1-2: Infrastructure Assessment

Conduct detailed assessment of existing data center facilities to determine available power, cooling, rack space, and network uplink capacity. Identify any facility upgrades required before new equipment installation can proceed. Document existing network architecture and identify integration points with current security infrastructure.

Week 2-3: Vendor Engagement and Procurement Initiation

Issue requests for quotation (RFQ) to preferred hardware vendors for server, storage, and networking equipment. Begin evaluation of colocation options if internal data center capacity is insufficient. Engage with network equipment vendors for proof-of-concept evaluations of proposed switching fabric.

Week 3-4: Team Formation and Training Planning

Identify personnel who will form the core platform engineering team. Assess skill gaps and initiate training plans for technologies new to the organization (Kubernetes/Docker

orchestration, Elasticsearch operations, Kafka administration). Establish development and staging environments for team onboarding and practice deployments.

Week 4-6: Detailed Design Finalization

Work with network architects to finalize IP addressing schemes, VLAN assignments, and routing policies. Document storage layout decisions including RAID configurations, LVM volume structures, and filesystem choices. Create detailed runbooks for initial deployment procedures that will be refined throughout the implementation process.

Week 6-8: Pilot Deployment

Execute pilot deployment using subset of hardware (minimum 3-4 servers) to validate deployment automation, integration procedures, and operational runbooks. Identify and resolve any issues discovered before full-scale deployment. Use pilot environment to train operations personnel on day-to-day administration tasks.

7.3 Success Metrics and Key Performance Indicators

Metric Category	Key Performance Indicator	Target	Measurement Method
Platform Availability	Overall system uptime	> 99.95%	Prometheus/Grafana monitoring
	Mean time to recovery (MTTR)	< 15 minutes	Incident ticket analysis
Event Processing	Events per second (sustained)	> 50,000 EPS	Kafka consumer lag metrics
	Event indexing latency (p99)	< 5 seconds	Elasticsearch index latency
	Alert generation latency	< 30 seconds	Wazuh alert timestamps
Investigation Efficiency	Search query response time (p95)	< 3 seconds	Kibana query logging
	Case resolution time (mean)	< 4 hours	TheHive case analytics
Data Integrity	Data loss events	Zero	Replication lag monitoring
	Backup success rate	100%	Backup job reports
Operational Readiness	Runbook coverage	> 90% procedures documented	Documentation audit
	Personnel certification	> 80% team certified	Training records

Table 7.1: Post-Deployment Success Metrics and Measurement Framework

This Hardware Architecture and Implementation Guide provides a comprehensive blueprint for deploying the Djezzy National SOC Platform in a production telecommunications environment. The recommendations contained herein represent

industry best practices adapted to the specific requirements of Algerian telecommunications operations, including data sovereignty considerations, regulatory compliance requirements, and the unique scale challenges posed by monitoring networks serving millions of subscribers. Successful implementation will establish Djazzy as a regional leader in telecommunications security operations, with capabilities matching or exceeding those of peer operators in European and Middle Eastern markets.